PHY201: Waves & Optics

Problem Set 2

Aug 20, 2025

1.
$$k = \frac{mg}{x} = \frac{2.5 \times 9.81}{0.06} = 408.75 \text{ N/m}.$$

For minimum settling time without overshoot, we choose critical damping: $\omega_0 = \gamma/2$ with $\omega_0 = \sqrt{k/m}$ and $\gamma = b/m$.

$$b_{\text{crit}} = 2\sqrt{km} = 2\sqrt{(408.75)(2.5)} \approx 6.393 \times 10^1 \text{ kg/s}.$$

$$b \approx 6.39 \times 10^1 \text{ kg/s}$$

2. For underdamped motion ($\gamma < 2\omega_0$),

$$Q = \frac{\omega_0}{\gamma}, \qquad \omega = \sqrt{\omega_0^2 - \frac{\gamma^2}{4}}.$$

- $\gamma = 0.01 \text{ s}^{-1}$:

$$Q = \frac{\pi}{0.01} \approx 3.14 \times 10^2, \quad \omega \approx \sqrt{\pi^2 - \tfrac{0.01^2}{4}} = 3.141589 \text{ rad/s}.$$

- $\gamma = 0.30 \text{ s}^{-1}$:

$$Q = \frac{\pi}{0.30} \approx 10.47, \quad \omega \approx \sqrt{\pi^2 - \tfrac{0.30^2}{4}} = 3.13801 \text{ rad/s}.$$

- $\gamma = 1.0 \text{ s}^{-1}$:

$$Q = \frac{\pi}{1.0} = 3.1416, \quad \omega \approx \sqrt{\pi^2 - \tfrac{1}{4}} = 3.10155 \text{ rad/s}.$$

Comment: Over this range, ω decreases only slightly.

(b) Plot (e.g. in a spreadsheet) for each γ:

$x(0) = 10$ mm, $\dot{x}(0) = 0$:

$$x(t) = e^{-\omega_0 t}(A + Bt), \quad A = x_0, \quad B = \omega_0 x_0.$$

Thus, with $x_0 = 10 \text{ mm} = 0.01$ m and $\omega_0 = \pi$,

$$x(t) = 0.01\,\big(1 + \pi t\big)e^{-\pi t} \text{ m}.$$

3. Given $x = A\sin\omega t$, the acceleration $a = \ddot{x} = -A\omega^2 \sin\omega t$, hence

$$P(t) = \frac{Ke^2a^2}{c^3} = \frac{Ke^2A^2\omega^4}{c^3}\sin^2\omega t.$$

Over one cycle $T = 2\pi/\omega$, $\langle \sin^2 \rangle = 1/2$, so

$$E_{\text{rad}} = \int_0^T P\,dt = \frac{Ke^2A^2\omega^4}{c^3}\frac{T}{2} = \frac{Ke^2\pi\omega^3A^2}{c^3}.$$

Total oscillator energy $E = \frac{1}{2}m\omega^2A^2$, so the quality factor

$$Q = \frac{2\pi E}{E_{\text{rad}}} = \frac{2\pi \cdot \frac{1}{2}m\omega^2A^2}{(Ke^2\pi\omega^3A^2/c^3)} = \frac{mc^3}{Ke^2\omega}.$$

For a visible photon, take $f \sim 5 \times 10^{14}\,\text{Hz}$, $\omega = 2\pi f \approx 3.14 \times 10^{15}\,\text{s}^{-1}$, and $e = 1.6 \times 10^{-19}$ C, $m = 9.1 \times 10^{-31}$ kg, $c = 3.0 \times 10^8$ m/s, $K = 6 \times 10^9\ \text{N}\,\text{m}^2\text{C}^{-2}$:

$$Q \approx \frac{(9.1 \times 10^{-31})(3.0 \times 10^8)^3}{(6 \times 10^9)(1.6 \times 10^{-19})^2(3.14 \times 10^{15})} \approx 5.1 \times 10^7.$$

Energy decays to 1/3 times its value in $t = 1/\gamma$. Thus, energy decay time $\tau \approx Q/\omega$,

$$\tau \approx \frac{5.1 \times 10^7}{3.14 \times 10^{15}}\ \text{s} \ \approx\ 1.6 \times 10^{-8}\ \text{s}\ \ (16\ \text{ns}).$$

4. Given $\bar{E}(t) = \bar{E}_0 e^{-\gamma t}$ and $Q = \omega_0/\gamma$. $\bar{E}(t) = \bar{E}_0/2$ at $t = 1$ s. Then $\gamma = \ln 2$.

   (a) Middle C: $f = 256$ Hz, $\omega_0 = 2\pi f = 512\pi$ rad/s.

   $$Q = \frac{512\pi}{\ln 2} \approx 2.32 \times 10^3.$$

   (b) One octave higher: $f = 512$ Hz, $\omega_0 = 1024\pi$ rad/s, same γ:

   $$Q = \frac{1024\pi}{\ln 2} \approx 4.64 \times 10^3.$$

   (c) $m = 0.1$ kg, $k = 0.9$ N/m $\Rightarrow \omega_0 = \sqrt{k/m} = \sqrt{9} = 3$ rad/s. Energy decays to $1/e$ in 4 s $\Rightarrow \gamma = 1/4 = 0.25\ \text{s}^{-1}$. Hence

   $$Q = \frac{\omega_0}{\gamma} = \frac{3}{0.25} = 12.$$

   Since $\gamma = b/m$, we have $b = m\gamma = 0.1 \times 0.25 = 0.025$ kg/s:

   $$b = 2.5 \times 10^{-2}\ \text{kg/s}.$$